

Qatar-1b

R-2661

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_190602 · created 2026-08-30T00:24:47+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458636.7851 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.056
Transit depth (Rp/Rs)^2	3.69 +/- 2.14 [%]
Orbital Inclination (inc)	88.58 +/- 2.89
Airmass coefficient 1 (a1)	1.98 +/- 0.53
Airmass coefficient 2 (a2)	-0.38 +/- 0.17
Scatter in the residuals of the lightcurve fit is	27.95 %
Best Comparison Star	None
Optimal Aperture	3.21
Optimal Annulus	7.5
Transit Duration (day)	0.0825 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.056 vs 0.1463 (deviation 0.82σ)
Duration fit vs published	0.0825 d vs 0.0692 d (deviation 1.69σ)
Mid-time O–C	0.9 min
Depth SNR	3.4
Residual scatter	27.95 %

Light curve

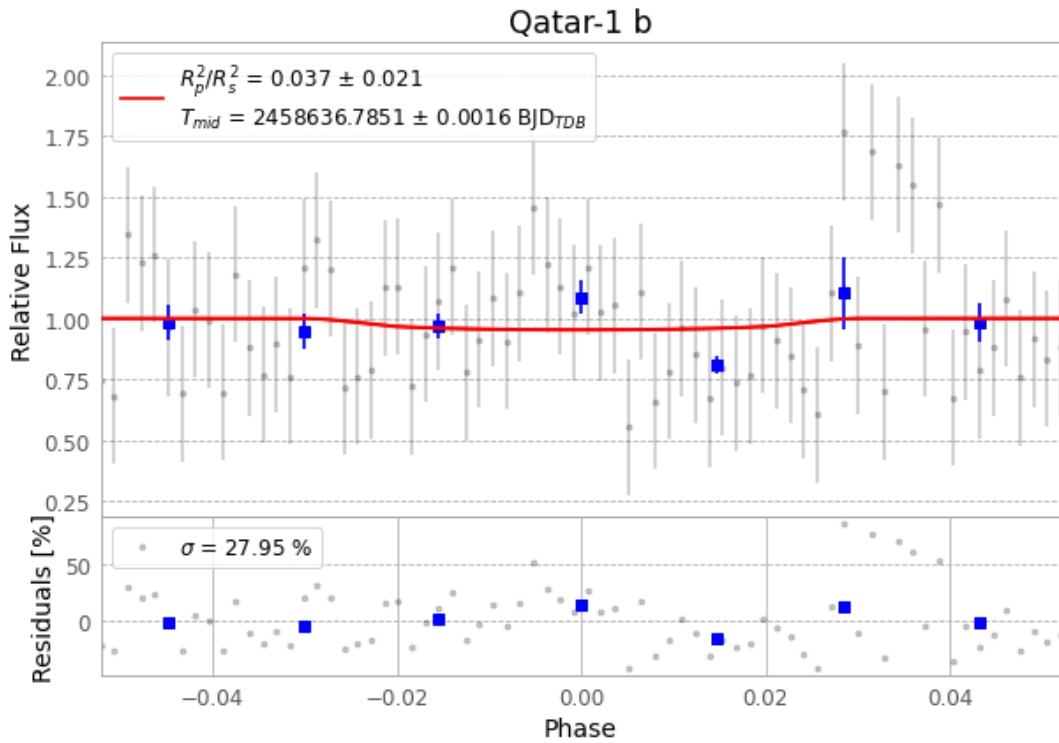


Figure 1 — FinalLightCurve_Qatar-1 b_2019-06-01.png (SHA-256 c37014c21b5b5bd5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 406], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 88a23c7e7649ceec...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1190602050312.FITS → Qatar-1190602083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-190602.log (072fe0039ded...), FinalLightCurve_Qatar-1 b_2019-06-01.png (c37014c21b5b...), FinalLightCurve_Qatar-1 b_2019-06-01.csv (43ece1914b45...)

Compute resources

Wall time 138.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 00:24Z.